Session Objectives

- 1. Describe and discuss the **general features and classifications of lung carcinoma** and evaluate its relationship to smoking
- 2. Discuss and compare the pathological features of **adenocarcinoma, squamous cell carcinoma, small cell carcinoma, and large cell carcinoma and correlate and describe their precursor lesions**
- 3. Discuss, describe, and distinguish some **genetic** aspects of the various types of **lung carcinomas**
- 4. Describe and explain the pathological features of **carcinoid tumor of the lung and metastatic nodules**
- 5. Discuss and describe the clinicopathological aspects, microscopic, and gross features of pleural neoplasms, **specifically malignant mesothelioma**

Topics

- General info/genetics
- Precursor lesions
- Adenocarcinoma (Non-small cell)
- Squamous cell carcinoma (Non-small cell)
- Small Cell carcinoma
- Large cell carcinoma
- Neuroendocrine tumors (Carcinoids)
- Metastatic tumors to lung
- Hamartoma
- Malignant mesothelioma

Lung Carcinoma

- Small Cell Carcinoma (15%)
- Non-small Cell Cancer [NSCLC]
  - –Squamous-cell carcinoma (20%)
  - –Adenocarcinoma (50%)
  - –Large-cell carcinoma (2%)
  - –Other/mixed (13%)

Features visible on light microscopy

Usually sufficient for most clinical needs

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Carcinoma of the Lung

- Most lung tumors (90 to 95%) are carcinomas
- Most frequently diagnosed major cancer in the world and most common cause of cancer mortality
- Older adults, 55 to 84 years old
- Major cause – cigarette smoking; other etiologies include radiation, asbestos (lung cancer and mesothelioma), radon
- 25% of lung cancers world wide are in non-smokers – tend to be women, tend to be adenocarcinomas – usually EGFR mutations
- Metastases common to liver (30 to 50%), brain (20%), bone (20%), and **adrenals (>50%)**

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Carcinoma of the Lung- Genetics

- Squamous cell and small cell especially associated with smoking

- Squamous cell shows **highest frequency of p53 mutations of all histological types, also**
  - CDKN2 (tumor suppressor gene) mutated in 65%
  - amplification of FGFR1

- **Small cell** also has frequent loss-of-function mutations involving **p53; RB loss-of-function; amplification of genes in MYC family**

- **Adenocarcinomas** may have gain-of-function mutations in multiple genes encoding receptor tyrosine kinases including **EGFR, ALK, ROS1, MET, RET, also:**
  - gain-of-function mutations in serine/threonine kinases such as BRAF, PIK3
  - KRAS gain-of-function mutations (30% of tumors)

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

WHY ARE GENETICS IMPORTANT?

- PROGNOSIS AND TREATMENT
- Patients with adenocarcinoma and activating EGFR mutations (15%) treated with specific kinase inhibitors which prolong survival
- However, KRAS mutations (30% of adenocarcinomas) usually have worse prognosis – although treatments on horizon

Robbins and Cotran, Pathologic Basis of Disease, 11th ed. 2025

Case question

A 53 year old woman with no history of smoking presents with a 3 month history of chest pain, cough, and mild fever. A chest Xray reveals a peripheral mass in the left upper lobe. When the mass is resected, the cut surface is mucinous and histological examination reveals well formed glands. What is the most likely diagnosis?

A. large cell carcinoma

B. mesothelioma

C. small cell carcinoma

D. adenocarcinoma

E. squamous cell carcinoma

Lung Adenocarcinoma

- Most common type in women and non-smokers
- Glandular differentiation/mucin production
- Grows more slowly than squamous cell Ca but tends to metastasize widely and earlier
- **KRAS mutations seen, esp. if smoker, correlate with worse outcome**
- **EGFR mutations and amplications (implication for inhibitor treatment)**

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Case question

A 52 year old woman has a one year history of truncal obesity and moderate depression. PE shows hirsutism and moon facies. Endocrinologic studies reveal hypokalemia, high plasma corticotropin levels, and increased serum and urine cortisol. CT scan of the thorax demonstrates a hilar mass. A transbronchial lung biopsy is performed (see next slide). EM shows neuroendocrine granules within the cytoplasm of some tumor cells. What is the most likely diagnosis?

A. Large cell carcinoma

B. Adenocarcinoma

C. Squamous cell carcinoma

D. Small cell carcinoma

E. Metastatic carcinoma

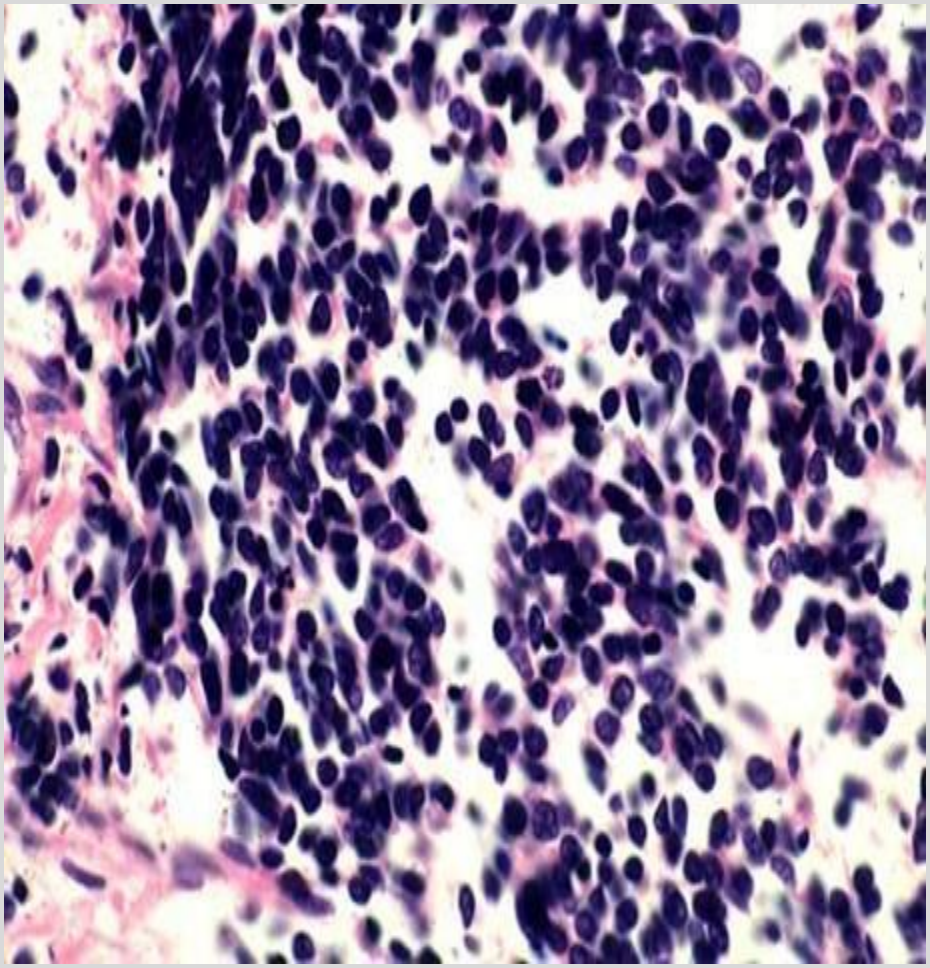
Small Cell Carcinoma

- Highly malignant, aggressive, 5-year survival rate very low
- **Neuroendocrine:**
  - histo shows small cells with scant cytoplasm, ill-defined cell borders
  - finely granular nuclear chromatin (“salt and pepper”)
  - absent/inconspicuous nucleoli
  - cells round to oval with nuclear molding
  - high mitotic rate
  - Azzopardi effect (basophilic staining of vascular walls due to encrustation by DNA from necrotic tumor cells)
- Strongest relationship to smoking
- **Dense-core neurosecretory granules (may be seen by EM) can secrete hormones ectopically;**

  + chromogranin,+synaptophysin
- **P53 and RB1 tumor suppressor genes frequently mutated**; high levels of BCL-2

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Small Cell carcinoma



- Small, blue cells with scant cytoplasm, absent or inconspicuous nucleoli, and finely granular nuclear chromatin
- Round, oval cells with nuclear molding
- High mitotic count, areas of necrosis
- +CK, +chromogranin, +synaptophysin

- EM shows neurosecretory granules – most likely to be associated with ectopic hormone production: **Cushing's syndrome (ACTH), SIADH - hyponatremia**

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Large Cell Carcinoma of lung

- Undifferentiated, malignant epithelial tumor which lacks cytology of small cell Ca and glandular or squamous differentiation; dx of exclusion
- Large nuclei, prominent nucleoli, moderate amount of cytoplasm
- May represent squamous cell or adenocarcinoma so undifferentiated that it is unrecognizable
- Do not confuse with large cell neuroendocrine carcinoma

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

LOCAL effects of LUNG CANCER

MOST COMMON COUGH, HEMOPTYSIS, CHEST PAIN, WEIGHT LOSS

Clinical Feature:	Pathologic Basis:
Pneumonia, abscess, lobar collapse	Tumor obstruction of airway
Lipid pneumonia	Tumor obstruction; accumulation of cellular lipid in foamy macrophages
Pleural effusion	Tumor spread into pleura
Hoarseness	Recurrent laryngeal nerve invasion
Dysphagia	Esophageal invasion
Diaphragm paralysis	Phrenic nerve invasion
Rib destruction	Chest wall invasion
SVC syndrome	SVC compression by tumor
Horner syndrome	Sympathetic ganglia invasion
Pericarditis, tamponade	Pericardial involvement

Robbins, 11th ed. 2025

SYSTEMIC effects of LUNG CANCER

(PARA-NEOPLASTIC SYNDROMES)~ 5%

ADH (hyponatremia) – ectopic production of hormone esp. in small cell carcinoma

ACTH (Cushing) - same

PTH-RP (Hyper-CA) - esp squamous cell CA

Other systemic manifestations

- Lambert-Eaton myasthenic syndrome – muscle weakness caused by autoAbs against neuronal calcium channel
- Peripheral neuropathy
- Trousseau syndrome (DVT and thromboembolism)
- Hypertrophic pulmonary osteoarthropathy – connective tissue abnormality – clubbing of fingers

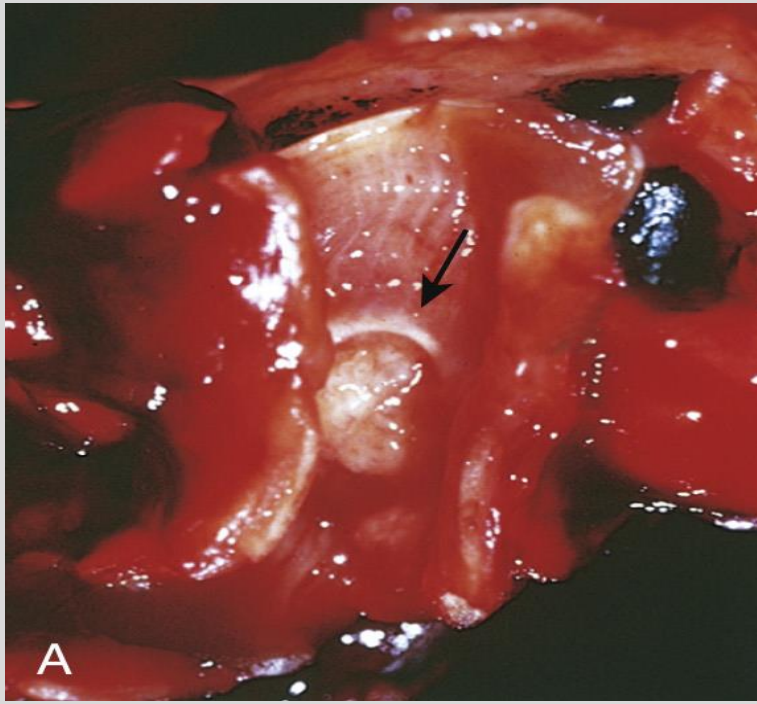
Robbins and Cotran, Pathologic Basis of Disease, 11th ed. 2025

NM Staging	
Tis	Carcinoma in situ Adenocarcinoma in situ: adenocarcinoma with pure lepidic pattern, ≤3 cm Squamous cell carcinoma in situ
T1	Tumor ≤3 cm without pleural or mainstem bronchus involvement (T1mi, minimally invasive adenocarcinoma; T1a, <1 cm; T1b, 1–2 cm; T1c, 2–3 cm)
T2	Tumor 3–5 cm or involvement of mainstem bronchus but not of carina, visceral pleural involvement, or lobar atelectasis (T2a, 3–4 cm; T2b, 4–5 cm)
T3	Tumor >5–7 cm or one with involvement of parietal pleura, chest wall (including superior sulcus tumors), diaphragm, phrenic nerve, mediastinal pleura, parietal pericardium, or separate tumor nodules in the same lobe
T4	Tumor >7 cm or any tumor with invasion of mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, or carina, or separate tumor nodules in a different ipsilateral lobe
N0	No metastasis to regional lymph nodes
N1	Ipsilateral intraparenchymal or peribronchial or hilar nodal involvement
N2	Metastasis to ipsilateral mediastinal or subcarinal lymph nodes
N3	Metastasis to contralateral mediastinal or hilar lymph nodes, ipsilateral or contralateral scalene, or supraclavicular lymph nodes
M0	No distant metastasis
M1	Distant metastasis (M1a, separate tumor nodule in contralateral lobe or pleural nodules or malignant pleural or pericardial effusion; M1b, single extrathoracic metastasis in a single organ; M1c, multiple extrathoracic metastases)

OF TECHNOLOGY
College of Osteopathic Medicine

“Carcinoid” Tumor of the Lung

- Low-grade malignant neoplasm of neuroendocrine cells (normal lung has NE cells within epithelium)
- Represent 1 – 5% of all lung tumors
- <60 y/o, ~20-40% non-smokers
- Gross appearance
  - ~3-4 cm
  - central tumor grows as a finger-like or spherical polypoid mass that often project into the mainstem bronchus
  - “Collar-button” lesion
  - Peripheral tumors are solid and nodular

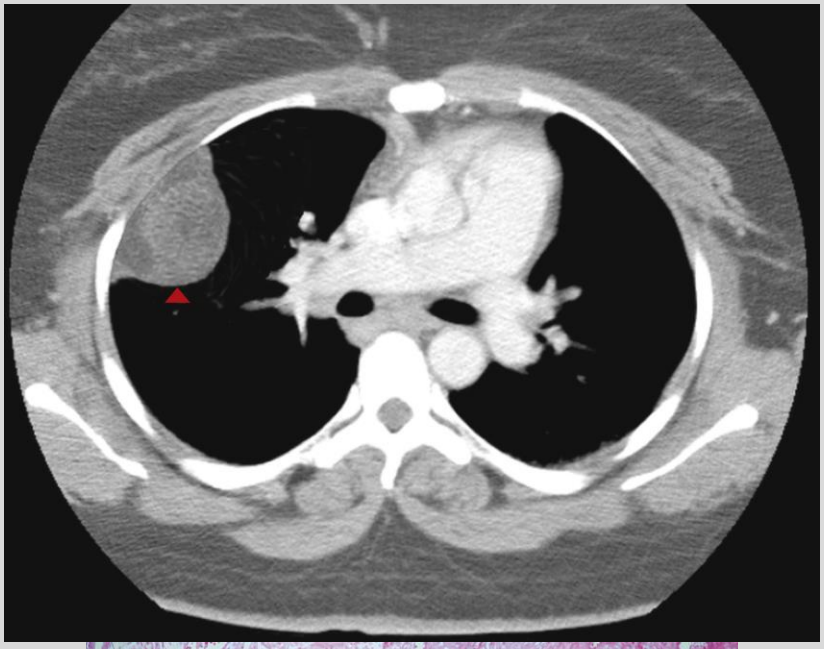


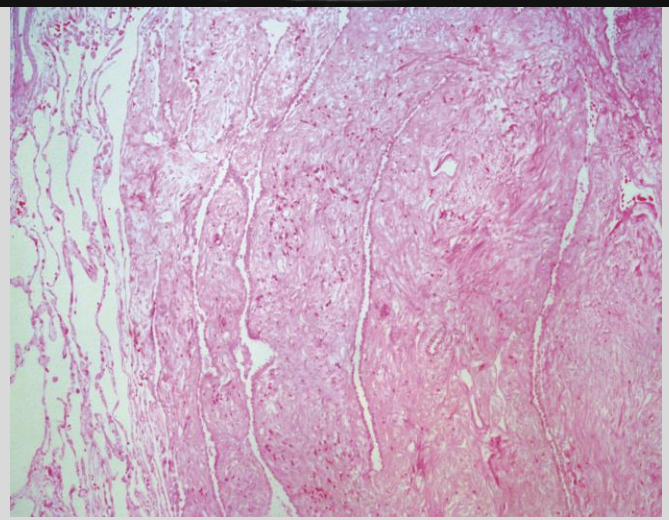
- Typical v. atypical
- about 10% produce carcinoid syndrome
  - Intermittent diarrhea, flushing, cyanosis

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Solitary Fibrous Tumors: Benign

- Soft tissue tumor common in the pleura (less commonly in lung)
- Unrelated to asbestos exposure
- Often attached to pleura by a pedicle
- Grossly it is composed of dense fibrous tissue with occasional cysts with viscous fluid
- often attached to pleural surface by pedicle (1 – 2 cm or > 10 cm)
- NAB2::STAT6 fusion gene unique to SFT
- Rarely may be malignant
- Whorls of reticulin and collagen
- Spindle cells resembling fibroblasts
- CD34+ and STAT6+
- Negative for keratin
  - Opposite of malignant mesothelioma





NEW YORK INSTITUTE OF TECHNOLOGY

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025

Robbins and Cotran, Atlas of Pathology, 9th ed 2021, Ch. 5

Case question

A 63 year old man who worked in a manufacturing plant for 30 years has had an 8 month history of chest discomfort, malaise, fever, night sweats, and weight loss. A chest Xray shows a pleural effusion and pleural mass encasing the lung. The patient subsequently dies of cardiorespiratory failure. Histological examination of the pleural mass at autopsy shows a biphasic pattern of epithelial and sarcomatous elements. What is the most likely diagnosis?

A. Carcinoid tumor

B. Small cell carcinoma

C. Metastatic carcinoma

D. Malignant mesothelioma

E. Large cell carcinoma

Malignant mesothelioma

- 90% asbestos related
- Lifetime risk in heavily exposed individuals 7 – 10%
- 80% have homozygous deletion of chromosome 9p leads to loss of tumor suppressor gene CDKN2A; also mutations in NF2, BAP1
- Epithelioid mesothelioma – 60%
- Sarcomatoid mesothelioma – 20%
- Mixed – 20%
- + calretinin, WT-1, cytokeratin 5/6, see previous slide
- 50% of patients die within 12 months of diagnosis
- Not just pleural – can arise in peritoneum and other sites

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

THE FOLLOWING 8 QUESTIONS COME FROM ACADEMIC SCHOLAR LOGAN MEUER CLASS OF 2024, ARKANSAS CAMPUS

Case Question 1

A 59 y/o female presents with a 6-month history of mild chest pain, cough, and low-grade fever. She has no significant PMH or family history. She does not smoke. Chest radiograph reveals a small, peripheral mass in the left upper lobe. What is the most likely diagnosis?

A. Small cell carcinoma
B. Large cell carcinoma
C. Adenocarcinoma
D. Squamous cell carcinoma
E. Mesothelioma

Case Question 2

What histological features will most likely be present in this patient's tumor?

A. Pleomorphic cells with no evidence of squamous or glandular differentiation
B. Keratinization with intracellular bridging
C. Small basophilic cells with necrotic areas
D. High degree of glandular differentiation with the presence of mucin

Case Question 3

A 65 y/o male presents with a 3-month history of persistent cough, chronic fatigue and weakness. He tells you, “I can barely make it up the stairs without stopping to catch my breath”. He has smoked about two packs per day for the last 30 years and consumes 1-2 alcoholic beverages per day. Chest x-ray reveals a centrally located mass near the right mainstem bronchus. What is the most likely diagnosis?

A. Small cell carcinoma
B. Large cell carcinoma
C. Adenocarcinoma
D. Squamous cell carcinoma
E. Mesothelioma

Case Question 5

A 54 y/o female presents with an 8-month history of chronic fatigue and mild depression. She has gained a significant amount of weight since her previous visit one year prior with her BMI increasing from 23 to 31. She has made no significant changes to her diet or lifestyle. PE reveals excessive hair growth on the face and neck as well as truncal obesity. CT scan of the thorax demonstrates a hilar mass. What histologic features will most likely be present in this patient's tumor?

A. Pleomorphic cells without squamous or glandular differentiation
B. Small, round basophilic cells with granular chromatin and high level of mitoses
C. Cells with uniform round nuclei and moderate cytoplasm
D. Glandular differentiation and mucin production
E. Squamous dysplasia with areas of focal necrosis

Case Question 7

A 50 y/o male presents with complaints of difficulty breathing that has persisted for about a week. He has no significant PMH or family history. PE reveals wheezing primarily on the right side close to the midline. Upon further questioning, he admits to having several watery stools over the last several days as well. He is otherwise in good health and has never smoked. CT scan of the chest reveals a 3 cm mass that projects into the right mainstem bronchus. What is the most likely diagnosis?

A. Mesothelioma
B. Large Cell Carcinoma
C. Carcinoid Tumor
D. Small Cell Carcinoma
E. Adenocarcinoma

Case Question 8

A 75 y/o retiree is watching television when he sees a commercial for a class action lawsuit for individuals exposed to asbestos who later developed mesothelioma. He believes he may be entitled to compensation and comes to your office to confirm that he has mesothelioma. If this patient were to be diagnosed with mesothelioma, what histological features would most likely be present in his tumor?

A. Mesenchymal type cells resembling a fibrosarcoma
B. Flattened cells forming tubular or papillary structures
C. Epithelial and mesenchymal tissue with no organizing pattern
D. Dense fibrous tissue with occasional cysts
E. Nodules of connective tissue interspersed with epithelial clefts

ANSWERS TO LAST 8 CASE QUESTIONS

- 1. C. Adenocarcinoma
- 2. D. High degree of glandular differentiation with the presence of mucin
- 3. D. Squamous cell carcinoma
- 4. A. Increased PTHrP: Elevated calcium on BMP
- 5. B. Cushing's syndrome secondary to small cell carcinoma of lung
- 6. A. p53 and RB
- 7. C. Carcinoid syndrome secondary to carcinoid tumor
- 8. B. Epithelioid type mesothelioma (60-80%)

Summary Slide

- Lung cancer- Classification and overview
- Adenocarcinoma - Peripheral, mucin production, TTF1 positive
  - Females, non-smokers
  - Atypical adenomatous hyperplasia (precursor lesion)
  - Adenocarcinoma-in-situ (formerly BAC), mucinous and non-mucinous
- Squamous cell carcinoma - Central, keratin, Correlated with smoking
  - p53 mutations
  - Hypercalcemia
- Small cell carcinoma - Very aggressive, highly malignant, highest degree of correlation with smoking
  - Neuroendocrine
  - P53 and RB tumor suppressor gene mutations
  - Small cells, high mitotic rate
  - ACTH-Cushing syndrome, ADH- SIADH leading to hyponatremia
- Large cell carcinoma – undifferentiated
- Carcinoid – low grade neuroendocrine neoplasm
- Hamartoma
- Metastatic cancer to lungs
- Pleural lesions – Pleural plaques, asbestos correlation, malignant mesothelioma (associated with asbestos exposure)